



The historical biogeography of the New World blackbirds (Icteridae)

Axel Arango¹, Chaitanya S. Gokhale¹

¹ Chair for Computational and Theoretical Biology, Klara-Oppenheimer-Weg 32, University of Würzburg, 97074, Würzburg, Germany

Corresponding authors: Axel Arango (axel.arango-garcia@uni-wuerzburg.de); Chaitanya S. Gokhale (chaitanya.gokhale@uni-wuerzburg.de)

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Abstract

The New World blackbirds (Icteridae) are a widely distributed, ecologically diverse avian family whose biogeographic history remains poorly understood. Competing hypotheses have proposed Neotropical or Nearctic origins, while the relative roles of dispersal, *in situ* speciation, and range evolution as drivers of diversification have remained unresolved. Here, the historical biogeography of Icteridae is reconstructed by integrating phylogenetic bioregionalization, probabilistic ancestral range estimation, biogeographic stochastic mapping (BSM), and Geographic Hidden State Speciation and Extinction (GeoHiSSE) modeling. Six phylogenetically distinct biogeographic regions were identified, and 18 models of range evolution were compared across three dispersal hypotheses. The best-supported model strongly supported a Nearctic origin, with subsequent stepwise southward colonization beginning with Mesoamerica (~7.53 Ma), followed by Amazonia (~6.06 Ma), the Andes (~5.84 Ma), Southern South America (~4.83 Ma), and finally Patagonia (~2.68 Ma). The earliest dispersal from North to South America predates the establishment of the Isthmus of Panama, coinciding instead with a documented dispersal wave of the Great American Biotic Interchange (GABI) and open-habitat expansion in northern South America. Dispersal was the dominant mode of range evolution, while *in situ* speciation was the primary cladogenetic process. The best-fit GeoHiSSE model revealed that geographic range alone does not fully explain diversification rate heterogeneity, with observed geographic states and unmeasured hidden rate classes jointly shaping macroevolutionary dynamics. Model-averaged speciation rates were comparable across regions. Together, these results support a Nearctic origin and stepwise southward colonization of the Americas by Icteridae, while demonstrating that lineage splitting is governed primarily by unmeasured biological factors within bioregions rather than geography *per se*.

Highlights

- Six distinct evolutionary regions were identified for Icteridae using phylogenetic bioregionalization.
- Strong support was found for a Nearctic origin of Icteridae, with subsequent stepwise southward colonization that predates the establishment of the Isthmus of Panama, rather than a South American origin, as proposed by Ernst Mayr in 1946.
- The expansion of Icteridae appears to be connected to the historical expansion of C4 grasses in the Americas.
- Speciation in Icteridae is not driven by geographical features; instead, it might be linked to other biological or ecological traits of the lineage, such as ecological specialization or sexual selection.
- Icteridae appears to follow the pattern of its parent clade (Emberizoidea) of within-region *in situ* speciation.

Keywords

bioregionalization, colonization, dispersal, GABI, Icteridae, Passeriformes, range evolution, speciation rates, SSE models

Introduction

The New World blackbirds (Icteridae) are among the most extensively studied avian clades in ecological, behavioral, and morphological research (Powell et al. 2014). This passerine family comprises approximately 106 extant species widely distributed throughout the Americas, from Alaska to Tierra del Fuego (Fraga 2008). Encompassing well-known groups such as New World orioles, grackles,

cowbirds, and meadowlarks, icterids exhibit remarkable diversity in evolutionarily important traits, including plumage, vocalizations, social behavior, and foraging strategies (Jaramillo and Burke 1999). As a result, members of this clade occupy nearly all major ecosystems in the New World, ranging from boreal forests and temperate grasslands to tropical rainforests, savannas, shrublands, and wetlands (Winkler et al. 2020). It is therefore reasonable to expect that their ecological and phenotypic diversity is closely tied to their biogeographic history, with the origin of the clade and subsequent dispersal across the Americas driving distinct distributional patterns and adaptive specializations in response to diverse environmental pressures (Ricklefs 2006a; Simões et al. 2016). Nevertheless, despite the prominence of the clade as a model system in ecological studies, its biogeographic history remains both controversial and understudied.

In 1946, Ernst Mayr theorized that the current distribution of Icteridae resulted from continuous colonization from South America, even before the Isthmus of Panama closed (Mayr 1946). Thus, despite being Pan-American, Icteridae was established as a clade of Neotropical origin. This hypothesis remained widely accepted until a recent study by Barker et al. (2015), which placed Icteridae decidedly as a clade of North American origin. Although this study provided robust and compelling evidence for a North American origin of Icteridae, its broad approach prevented a more detailed understanding of the biogeographic history of the clade (Powell et al. 2014). As a result, key evolutionary processes, such as within-continental dispersal and *in situ* diversification, remain unresolved (Daru et al. 2017). For instance, it is still unclear whether the expansion of the group followed a stepwise pattern through adjacent regions, as Mayr originally proposed, or whether dispersal was more opportunistic and unconstrained, consistent with the high dispersal ability and migratory behavior of many icterid species (Kondo and Omland 2007; Arango et al. 2022). Similarly, the extent to which these dynamics and geographic barriers have shaped diversification across the Americas remains highly debated (Ricklefs 2006b; Machac 2020).

To better understand biodiversity patterns, it is crucial to define biogeographic boundaries that reflect historical processes (Olson et al. 2001; Kreft and Jetz 2010). Incorporating evolutionary information into boundary delimitation reveals both the spatial and evolutionary structure of species distributions (Daru et al. 2016, 2017; Kennedy et al. 2017) and sheds light on the mechanisms driving these patterns (Leprieur et al. 2012; Daru et al. 2017). Therefore, by identifying clusters of lineages that diversified within shared biogeographic contexts, evolutionarily informed bioregionalizations offer insights into spatial aspects of evolutionary history, such as dispersal and *in situ* diversification (Baselga and Orme 2012; Daru et al. 2017). This perspective is particularly important for clades such as Icteridae, which exhibit high ecological versatility and broad distributions across heterogeneous landscapes (Winkler et al. 2020).

As such, incorporating these evolutionarily informed bioregions into parametric models of range evolution can

deepen the understanding of how geography and range evolution have shaped the diversification of lineages (Matzke 2013). This approach provides a more comprehensive view of the patterns, processes, and historical events that underlie present-day biodiversity distributions (Leprieur et al. 2012; Matzke 2014; Arango et al. 2025; Lima et al. 2025). Moreover, geography can also influence macroevolutionary dynamics by mediating speciation and extinction rates across regions that might differ in area, connectivity, ecological heterogeneity, and isolation (Goldberg et al. 2011; Rolland et al. 2014; Beaulieu and O'Meara 2016). In the case of Icteridae, North and South America could differ markedly in their biogeographic histories, climatic regimes, and habitat diversity, with differential diversification rates between the two regions further highlighting the main processes shaping the biodiversity of the clade (Ricklefs 2006b; Smith et al. 2014).

In this study, the historical biogeography of Icteridae is reconstructed, and the influence of geography on its diversification is assessed by integrating phylogenetic bioregionalization with comparative modeling. First, evolutionarily coherent bioregions across the New World are delineated using a Phylogenetic Bioregionalization framework (Daru et al. 2020), providing biologically meaningful units for ancestral range estimation with BioGeoBEARS (Matzke 2014). The most common models of range evolution are compared across three different hypotheses: an unconstrained model with equal dispersal among regions, a dispersal-restricted model incorporating geographic connectivity, and a directional model allowing asymmetric movement between North and South America. Biogeographic stochastic mapping (BSM) is then used to reconstruct the biogeographic events of the clade. To evaluate the impact of geography on diversification rates, Geographic Hidden State Speciation and Extinction (GeoHiSSE) models (Beaulieu and O'Meara 2016) are applied to estimate speciation and extinction across regions. Together, these approaches clarify how geography, dispersal, and regional dynamics have shaped the evolutionary history of Icteridae.

Methods

Geographic and phylogenetic data

The geographic distributions of icterid species were obtained from the BirdLife International range maps v. 2024 (BirdLife International and Handbook of the Birds of the world 2024), comprising 104 of the 106 species recognized as of 2025 according to the Clements Checklist (Clements et al. 2025). The phylogenetic relationships of these 104 species were obtained by pruning the Icteridae clade from the maximum clade credibility (MCC) phylogeny of Barker et al. (2015). Until recently, *Sturnella lilianae* was lumped with *S. magna* and therefore has no distinct spatial data available. *Cacicus microrhynchus* was excluded because of the absence of phylogenetic data for this species in the Barker et al. (2015) tree used for the analyses.

Phylogenetic biogeographical regionalization

To reconstruct the biogeographic history of Icteridae, the pertinent biogeographic regions for the clade were first delimited. To delimit these regions, a phylogenetic bioregionalization was applied. This method identifies evolutionarily distinct biogeographic regions by taking into consideration the evolutionary relatedness of species in a geographic context (Daru et al. 2017).

To perform the bioregionalization, the breeding distribution maps of Icteridae species were first transformed into a $1 \times 1^\circ$ presence–absence matrix. Using this matrix, phylogenetic turnover between assemblages was calculated with Simpson's Phylogenetic Index of dissimilarity (pb_{sim}). This index measures turnover based on shared branch lengths among grid cells and is particularly suited for biogeographic applications because of its insensitivity to differences in species richness (Kreft and Jetz 2013; Daru et al. 2016; Gross et al. 2025). Breeding-season range maps were used for all analyses, following standard practice in biogeographic reconstruction (Belmaker et al. 2012; Jetz et al. 2012; Kennedy et al. 2017; Cai et al. 2021), as these distributions might more closely reflect the realized niche of the species (Matthiopoulos 2022). In the case of Icteridae, this is further supported because migration within the clade is thought to have evolved primarily through the addition of non-breeding ranges onto a conserved breeding distribution (Winger et al. 2014).

Several hierarchical clustering methods were then evaluated to determine which best captured the structure of the phylogenetic dissimilarity matrix. These included complete linkage, UPGMA (unweighted pair-group method using averages), UPGMC (using centroids), WPGMA (weighted averages), WPGMC (weighted centroids), Ward's minimum variance, and the divisive method DIANA. Model performance was assessed using the cophenetic correlation coefficient (Sokal and Rohlf 1962).

To determine the optimal number of clusters (k), the “elbow” method (Salvador and Chan 2004) was applied, which identifies the point beyond which adding more clusters yields increases in explained variance of no more than 0.01 (Daru et al. 2020). While cluster selection is often considered arbitrary (Kreft and Jetz 2013), the elbow method provides a more objective criterion by identifying the point at which the explained variance of the phylogenetic matrix levels off (Salvador and Chan 2004).

Finally, non-metric multidimensional scaling (NMDS) was performed based on the distance matrix derived from the best-performing clustering algorithm (UPGMA) and the optimal number of clusters. The result was a set of spatially coherent, phylogenetically distinct biogeographic regions for the Icteridae family.

Biogeographic modeling

The previously delimited biogeographic regionalization was used to evaluate distinct models of ancestral area reconstruction using the BioGeoBEARS R package (Matzke

2014) and to reconstruct the historical range evolution of Icteridae. BioGeoBEARS allows for the comparative evaluation of multiple likelihood-based biogeographic models within a common statistical framework (Matzke 2013).

Three widely used models of range evolution (DEC, DIVALIKE, and BAYAREALIKE) were tested, each with and without the addition of the founder-event speciation parameter (+J), resulting in a total of six candidate models per hypothesis. The models were informed by a discrete biogeographic dataset based on the results of the phylogenetic bioregionalization, with each species assigned to one or more of the identified bioregions.

To account for the geographic structure and varying connectivity between regions, a geodesic distance matrix representing pairwise great-circle distances between the boundaries of each bioregion was incorporated (Suppl. material 1: table S1). Furthermore, different dispersal matrices were used to modify the dispersal probability multipliers between regions under different hypotheses of range evolution (Suppl. material 1: table S2). Specifically, three hypotheses were tested: (A) an unconstrained dispersal model in which all regions were assumed to have equal pairwise dispersal probabilities; (B) a dispersal-restricted model in which probabilities were inversely scaled by geographic distance, reflecting reduced dispersal likelihood over longer distances; and (C) a directional dispersal model in which asymmetries in dispersal rates were introduced to simulate biased movement from North to South America, consistent with a hypothesized southward expansion of the clade.

Each model was fitted to the empirical range data and the maximum clade credibility tree using likelihood optimization. Model fit was evaluated using the log-likelihood ($\ln L$) and corrected Akaike information criterion (AICc) scores. Comparisons among models were used to identify the best-fitting biogeographic scenario and assess the relative importance of dispersal constraints and founder-event speciation in shaping the distribution of Icteridae.

To test which dispersal hypothesis models best explain icteric biogeographic events, likelihood ratio tests were performed between the best models of each hypothesis. Following model selection, biogeographic stochastic mapping (BSM) was implemented under the best-supported model to estimate the number, direction, and timing of historical dispersal, extinction, and range-switching events. A total of 1000 stochastic maps were generated to summarize the frequency and spatial pattern of inferred events across the phylogeny, providing a probabilistic view of the processes that have shaped the current distribution of Icteridae.

Geographic diversification

To evaluate the role of geography in shaping diversification rates within Icteridae, a series of geography-dependent diversification models was implemented using the GeoHiSSE framework (Caetano et al. 2018). To do so, species were classified as endemic to North America (NA), endemic to South America (SA), or widespread (WS).

Four classes of models were compared to test whether geographic occupancy is associated with differences in speciation, extinction, or transition rates. First, a null model was fitted in which diversification is independent of geography, assuming equal rates across continents. Second, the canonical GeoSSE model was tested, in which speciation, extinction, and range transition rates are allowed to differ between lineages restricted to different regions and those occupying multiple areas. Third, a null model with hidden states was applied that maintains geographic independence while allowing diversification rates to vary across unobserved (hidden) character states, thereby accounting for potential rate heterogeneity unrelated to geography. Finally, a set of GeoHiSSE models was tested that incorporates both observed geographic states and hidden rate classes, allowing for complex diversification dynamics structured by geography and unmeasured biological or ecological factors.

By comparing the fit of these models using the Akaike information criterion (AIC), whether the observed geographic distribution of lineages is associated with shifts in diversification rates was assessed while controlling for background rate heterogeneity. This approach provides a robust framework to evaluate whether regional occupancy has had a significant impact on the diversification of Icteridae. Furthermore, to evaluate the contribution of each region to the speciation of Icteridae, the resulting parameters of each model at the tips were averaged, weighted by their AIC weights (AICw) using a harmonic mean, as suggested by Caetano et al. (2018).

Results

Phylogenetic biogeographical regionalization

The best-performing clustering algorithm for Icteridae phylogenetic turnover was UPGMA (Suppl. material 1: table S3). According to the “elbow” method, the optimal number of regions (k) was 6, based on convergence, a low level of stress, and explained variance (Suppl. material 1: fig. S1, table S4).

Accordingly, the NMDS ordination defined six phylogenetic biogeographic regions for Icteridae (Suppl. material 1: fig. S2): Nearctic (A), Andes (B), Amazonia (C), Southern South America (D), Patagonia (E), and Mesoamerican-Caribbean (F; Fig. 1a).

Biogeographic modeling

Model comparisons revealed that the best-fitting model under the unconstrained dispersal hypothesis was BAYAREALIKE, which assumed widespread sympatry and frequent range expansions without penalizing geographic distance. Under both the adjacent-area dispersal and gradual directional dispersal hypotheses, DEC and DEC+J models performed similarly (Table 1).

Given that the best-fitting model for the unconstrained hypothesis was largely outperformed by the best-fitting models for both the adjacent and gradual hypotheses, it was excluded from the likelihood ratio test. Furthermore, since the likelihood ratio test showed no difference between DEC and DEC+J under either the adjacent or gradual hypothesis, the models with fewer parameters (DEC) were used to represent each hypothesis in the final comparison. The likelihood ratio test comparing the best adjacent-area model (DEC) and the best gradual model (DEC) showed that the gradual model provided a significantly better fit (Suppl. material 1: table S5).

Ancestral area reconstruction under the best-fit gradual DEC model strongly supported a Nearctic origin for the clade, with the highest marginal probability concentrated in bioregion A (Fig. 1b). Accordingly, BSM under the gradual DEC model revealed strong support for directional dispersal from the ancestral area of Icteridae in the Nearctic into Neotropical regions.

Across 1000 stochastic maps, the main mode of range evolution for Icteridae was dispersal, with an average of 161.4 ± 5.9 anagenetic dispersal events, whereas range contraction was significantly lower (32.9 ± 4.58). Most cladogenetic events in the clade were *in situ* events, with an average of 55.25 ± 4.49 , followed by range-expansion cladogenesis, with an average of 30.42 ± 4.58 events. Although rarer, vicariant events were also present, with an average of 16.32 ± 3.2 . These results indicate that the main mode of cladogenesis for Icteridae was within-region speciation (Suppl. material 1: table S6, fig. S3).

BSM reconstructed the origin of Icteridae in the Nearctic region (A), from which the clade subsequently expanded across the Americas in a southward progression. The BSM results showed that the earliest dispersal event occurred approximately 7.53 ± 1.35 Ma, when lineages colonized the Mesoamerican-Caribbean region (F). The Amazonia region (C) and the Andes region (B) were reached thereafter, approximately 6.06 ± 1.15 Ma and 5.84 ± 1.46 Ma, respectively. Colonization continued southward, with lineages establishing in Southern South America (D) approximately 4.83 ± 1.28 Ma and ultimately reaching the southernmost extent of the range of the family, the Patagonian region (E), approximately 2.68 ± 1.63 Ma (Fig. 2a; Suppl. material 1: table S7).

BSM revealed a clear directionality in the dispersal history of Icteridae, consistent with a Nearctic origin and progressive southward colonization of the Neotropics. The highest dispersal flow out of the Nearctic was directed toward Mesoamerica (22.58 ± 5.04 events), which itself functioned as a major stepping-stone toward South America, dispersing extensively into both the Andes (18.16 ± 3.28) and Amazonia (15.97 ± 3.18). From the Andes, lineages continued to spread southward, with dispersal toward Amazonia (11.93 ± 3.84) and Southern South America (11.1 ± 2.84) exceeding the reverse directions (10.05 ± 3.2 and 4.15 ± 1.9 , respectively), reinforcing a net south-directed dynamic. Amazonia further contributed substantially to the colonization of Southern South America (22.45 ± 4.76 vs. 6.61 ± 3.41 events in

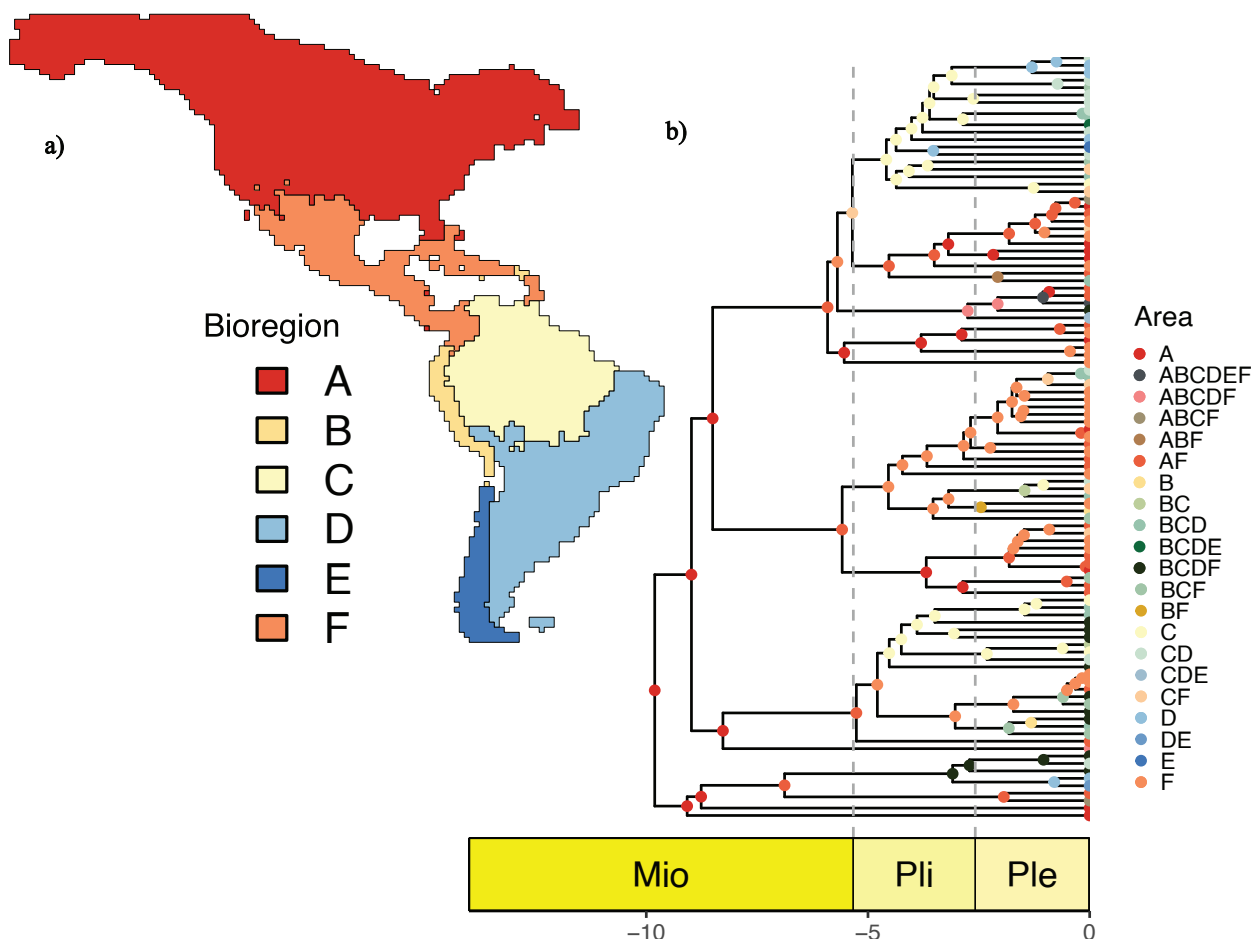


Figure 1. Biogeographic regionalization and ancestral area reconstruction for the New World blackbirds (Icteridae; $n = 104$ species). **a.** Six phylogenetically distinct bioregions delimited across the Americas via non-metric multidimensional scaling (NMDS) of phylogenetic turnover: Nearctic (A); Andes (B); Amazonia (C); Southern South America (D); Patagonia (E); Mesoamerican-Caribbean (F); **b.** Most probable ancestral state estimates under the gradual DEC model (dispersal = 0.37, extinction = 0.09, log-likelihood [$\ln L$] = -284.2) for Icteridae. Colors with single letters correspond to the bioregions in **a**), while colors with multiple letters represent an area resulting from a combination of these bioregions (e.g., AF represents a region formed by the Nearctic and Mesoamerican-Caribbean).

the opposite direction), representing the final major step in this continental progression. Patagonia emerged as a distal peripheral sink at the terminus of this colonization axis, receiving dispersal almost exclusively from Southern South America (6.72 ± 1.28), with negligible backflow, and contributing little to outward dispersal across any route. Overall, the Nearctic, Mesoamerica, and Amazonia were the dominant net source regions (net = +7.95, +11.31, and +6.87 events, respectively), while Southern South America (-17.51) and Patagonia (-6.11) functioned as net recipients, tracing a coherent south-directed colonization pathway from North America through the Isthmus into the heart of South America (Suppl. material 1: tables S8, S9; Fig. 2b).

Range contraction events (Suppl. material 1: table S10), reflecting local extinction or range shrinkage within areas, were also unevenly distributed across regions. Mesoamerica exhibited the highest mean number of range contractions (mean = 8.1 ± 2.62), exceeding all other areas and suggesting high lineage turnover despite its role as a major dispersal corridor and center of *in situ* diversification.

The Nearctic (6.82 ± 2.33), Andes (6.75 ± 2.54), and Amazonia (5.29 ± 2.16) showed intermediate and similar contraction counts, indicating comparable levels of range loss across these core areas. Southern South America had somewhat fewer contractions (4.2 ± 1.92), while Patagonia showed markedly lower range contraction rates (1.78 ± 1.27), consistent with its very low lineage diversity and limited overall involvement in Icteridae biogeographic dynamics. Across all areas, most extirpation events occurred from 2.5 Ma onward, with a strong bias toward the present (Suppl. material 1: fig. S4).

In situ speciation events were unevenly distributed across the six biogeographic areas, reflecting both the antiquity and the diversification potential of each region (Suppl. material 1: table S11). Mesoamerica (F) showed the highest mean number of within-area speciation events (mean = 22.3 ± 4.7), followed by Amazonia (C; 13.3 ± 5.47) and the Nearctic (A; 12.8 ± 4.22), suggesting that these regions not only served as sources of dispersal but also as important centers of *in situ* diversification. Southern South

Table 1. Model selection results from BioGeoBEARS for 104 species of Icteridae (New World blackbirds), comparing 18 candidate models of range evolution across three dispersal hypotheses (unconstrained, adjacent-area, and gradual). $\ln L$ = log-likelihood; n = number of parameters; d = dispersal rate; e = extirpation (or local extinction) rate; j = jump dispersal rate. Best-fit models for each hypothesis are highlighted in grey.

Hypothesis	Model	$\ln L$	n	d	e	j	AIC
Unconstrained	DEC	-360.6	2	0.074	0.061	0	725.22
	DEC+J	-358.1	3	0.097	0.214	1×10^{-5}	722.16
	DIVALIKE	-384.1	2	0.075	1×10^{-12}	0	772.18
	DIVALIKE+J	-381.9	3	0.072	1.51×10^{-8}	0.01	769.76
	BAYAREALIKE	-332.5	2	0.034	0.195	0	669.05
	BAYAREALIKE+J	-327.4	3	0.030	0.141	0.013	660.72
Adjacent	DEC	-288.7	2	0.182	0.060	0	581.34
	DEC+J	-288.7	3	0.186	0.062	1×10^{-5}	583.32
	DIVALIKE	-298.5	2	0.189	1×10^{-12}	0	601.00
	DIVALIKE+J	-298.5	3	0.189	1×10^{-12}	1×10^{-5}	603.00
	BAYAREALIKE	-319.0	2	0.057	0.096	0	641.94
	BAYAREALIKE+J	-292.0	3	0.176	0.250	0.016	589.97
Gradual	DEC	-284.6	2	0.377	0.093	0	573.2
	DEC+J	-284.6	3	0.376	0.094	1×10^{-5}	575.2
	DIVALIKE	-297.1	2	0.355	1×10^{-12}	0	598.1
	DIVALIKE+J	-297.1	3	0.355	1×10^{-12}	1×10^{-5}	600.1
	BAYAREALIKE	-292.2	2	0.301	0.234	0	588.4
	BAYAREALIKE+J	-288.3	3	0.289	0.181	0.104	582.5

America (D; 4.56 ± 4.03) and the Andes (B; 2.26 ± 2.43) showed intermediate speciation counts, though with notably high variance, possibly reflecting heterogeneity in diversification dynamics across the distribution of stochastic maps. Patagonia (E) stood in stark contrast to all other regions, with a nearly absent *in situ* speciation signal (mean = 0.004 ± 0.06), consistent with its role as a peripheral sink colonized too recently or too sporadically to have accumulated substantial *in situ* diversification.

In terms of timing, *in situ* speciation events in the Nearctic were, on average, the oldest (mean = 4.81 ± 2.93 Ma), indicating that diversification within this ancestral area began earlier than in any other region and predates the major southward colonization pulses. Speciation events in Southern South America (2.51 ± 1.32 Ma) were somewhat more recent on average, consistent with a later arrival of lineages at the southern end of the colonization axis. The remaining regions—the Andes (3.18 ± 1.19 Ma), Amazonia (2.98 ± 1.32 Ma), and Mesoamerica (2.06 ± 1.60 Ma)—showed similar and relatively recent mean speciation times, suggesting a roughly contemporaneous burst of *in situ* diversification across the Neotropics during the Pliocene–Pleistocene (Suppl. material 1: table S12; Fig. 2c). In Patagonia, only four stochastic maps recovered an *in situ* event, all at the common ancestor of *Amblyramphus holosericeus* and *Curaeus curaeus*, approximately 3.35 Ma.

Geographic diversification

Model comparison across GeoSSE, GeoHiSSE, and null models revealed that the full GeoHiSSE model provided the best fit to the data (Table 2), indicating that

diversification dynamics in Icteridae are shaped by both observed geographic occupancy and unmeasured hidden states. This result suggests that geography alone does not fully capture the heterogeneity in speciation and extinction rates across the clade and that at least one additional biological or ecological factor (not explicitly modeled here) contributes to diversification rate variation among lineages.

The two hidden rate classes reveal distinct macroevolutionary regimes. In hidden rate class A, turnover is low to moderate and geographically structured, with SA endemics showing the lowest values ($\tau = 0.05$ sp/My) and WS lineages the highest ($\tau = 0.48$ sp/My), followed by NA lineages ($\tau = 0.45$ sp/My). All ranges had

Table 2. Model selection of state-dependent speciation and extinction (SSE) models. Null is a model in which diversification is independent of geography; GeoSSE is the canonical GeoSSE model in which diversification is dependent on geography; Hidden Null is a model in which diversification is independent of geography but related to unmeasured traits; and GeoHiSSE is a model in which diversification is affected by both geography and an unmeasured trait. $\ln L$ = log-likelihood; AICc = corrected Akaike information criterion; AICw = Akaike weight, representing the relative support for each model among those compared. NA = North America endemic; SA = South America endemic; WS = widespread (occurring in both regions).

Model	$\ln L$	AICc	AICw
Null	-289.86	585.96	0.04
GeoSSE	-288.53	587.68	0.07
Hidden Null	-286	582.61	0.051
GeoHiSSE	-277.09	579.09	0.936

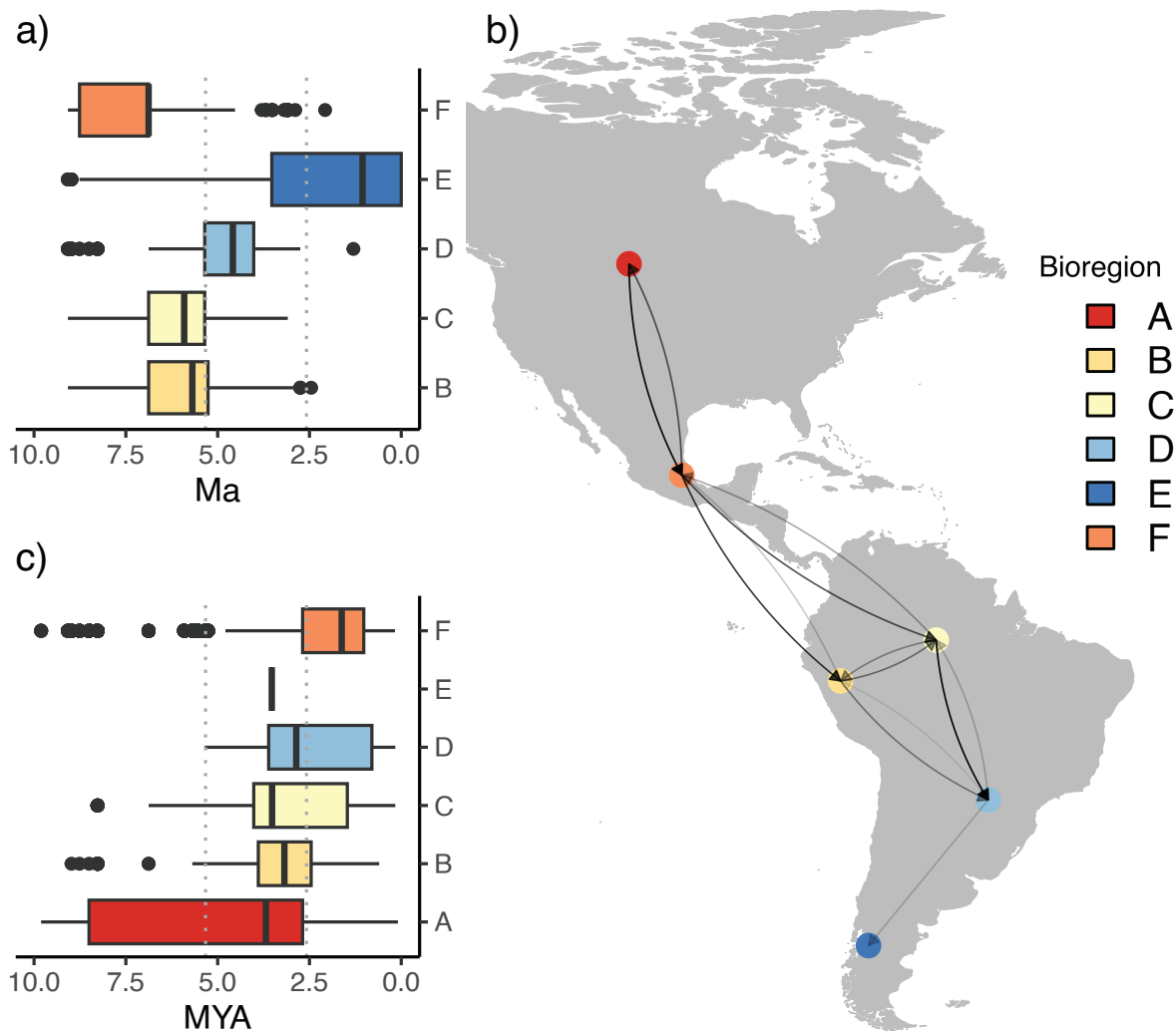


Figure 2. Timing, direction, and intensity of range evolution events for Icteridae, summarized across 1000 biogeographic stochastic maps (BSMs). **a.** Timing of first colonization events for each region (except for ancestral region A) according to the 1000 BSMs: B (median = 5.69 Ma [IQR = 1.62]); C (median = 5.91 Ma [IQR = 1.53]); D (median = 4.59 Ma [IQR = 1.33]); E (median = 1.05 Ma [IQR = 3.53]); F (median = 6.88 Ma [IQR = 1.88]); **b.** Dispersal direction and intensity between regions according to the mean number of dispersal events per region, averaged across the 1000 BSMs; bolder arrows indicate higher intensity (Suppl. material 1: tables S8, S9); **c.** Distribution of the timing of *in situ* speciation events by region, calculated from the 1000 BSMs: A (median = 3.69 Ma [IQR = 5.58]); B (median = 3.18 Ma [IQR = 1.44]); C (median = 3.52 Ma [IQR = 2.55]); D (median = 2.86 Ma [IQR = 2.81]); E (median = 3.53 Ma [IQR = 0]); F (median = 1.62 Ma [IQR = 1.67])

negligible extinction fractions. In hidden rate class B, the geographic structure shows a completely different pattern. NA endemics show near-zero turnover, SA endemics have moderate values ($\tau = 0.46$ sp/My), while WS lineages show extraordinarily high values ($\tau = 11.73$ sp/My). Since extinction fractions were negligible across endemic states, all rate differences are driven by variation in speciation rather than differential extinction. In the case of WS lineages, the extinction fraction cannot be calculated because, for a lineage to go extinct, its range first needs to contract (become endemic). Because of this caveat, turnover in WS lineages represents the sum of three cladogenetic speciation components (within-NA, within-SA, and between-area vicariant speciation; Caetano et al. 2018), not a combination of speciation and extinction. The high turnover therefore reflects an elevated

overall rate of cladogenetic events across all three ranges simultaneously rather than high lineage turnover in the conventional sense. Concordantly, speciation rates averaged across all models were highest in WS lineages (0.334 sp/My), followed by SA endemics (0.314 sp/My), with both having similar rates of speciation, and NA endemics (0.234 sp/My), which showed slightly lower rates. This reflects the highly transitory nature of WS lineages and the combined contribution of both hidden rate classes weighted by their marginal probabilities across the phylogeny (Table 3).

These results are congruent with a scenario in which Icteridae originated in the Nearctic and diversified primarily *in situ* within newly colonized regions, with sustained diversification in both North and South America over evolutionary time (Fig. 3).

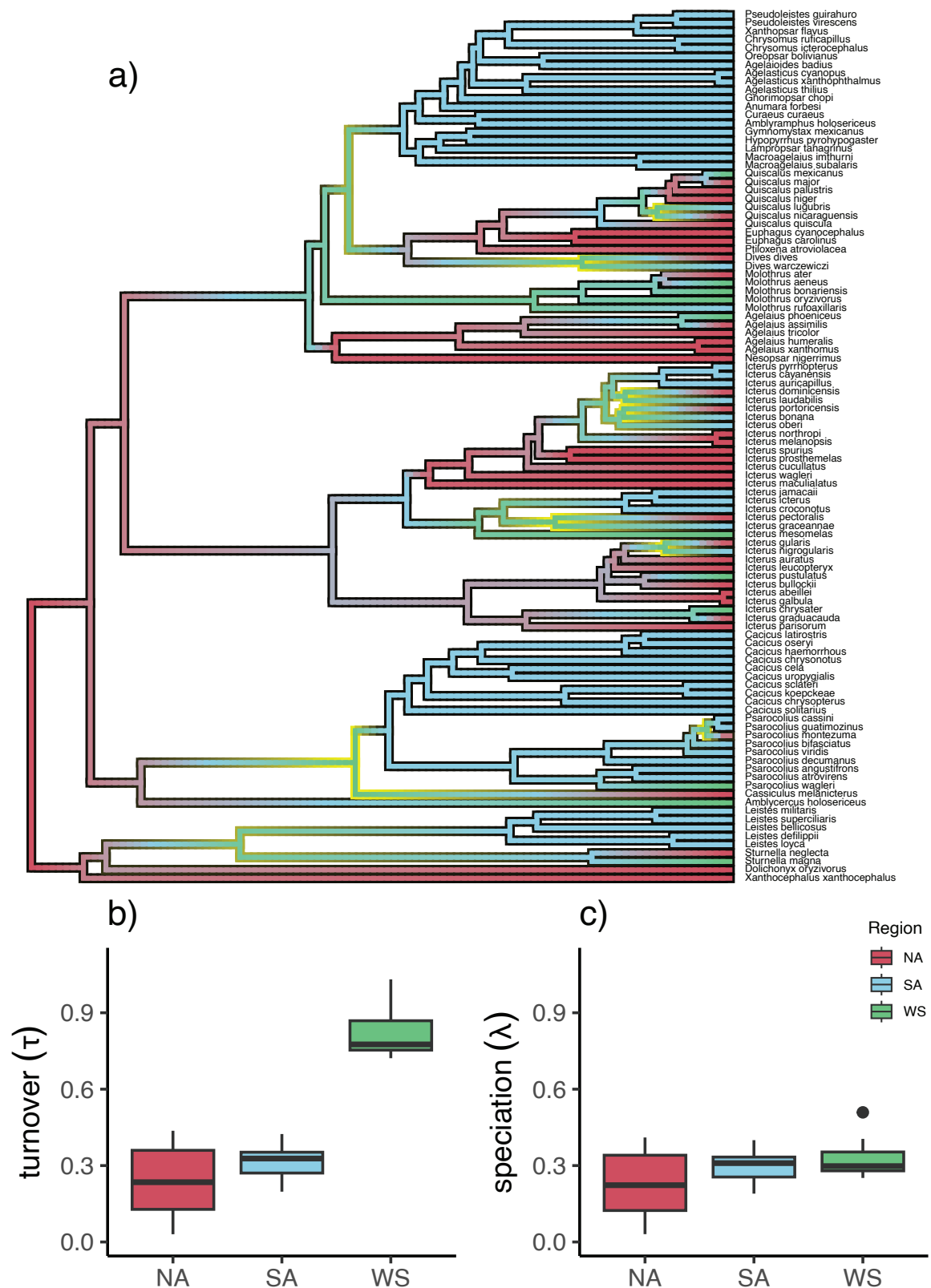


Figure 3. Model-averaged marginal ancestral state reconstruction of geographic occupancy and associated cladogenetic rates across the Icteridae phylogeny (104 species). **a.** Marginal reconstruction of geographic states mapped onto the Icteridae phylogeny, with branch colors indicating the most probable geographic state at each node and tip: red = North America endemic, blue = South America endemic, and green = widespread (present in both regions). Branch color intensity reflects the marginal probability of the most probable state, with more saturated colors indicating higher confidence. Yellow branch outlines indicate lineages with higher model-averaged speciation rates; **b.** Distribution of model-averaged turnover rates (τ) across tips grouped by geographic state (NA = North America endemic, SA = South America endemic, WS = widespread). **c.** Distribution of model-averaged speciation rates (λ) across tips grouped by geographic state.

Table 3. Parameter estimates from the best-fit GeoHiSSE model (AICw = 0.936) for Icteridae. For each combination of geographic state and hidden rate class, turnover (τ = speciation + extinction) and extinction fraction (ϵ = extinction/speciation) are reported from the maximum likelihood solution. The speciation column (λ) reports model-averaged rates across all candidate models at the tips of the phylogeny, weighted by AICc weights, and represents the primary summary statistic for geographic diversification differences. NA = North America endemic; SA = South America endemic; WS = widespread. Both hidden rate classes (A and B) are unobserved and unmeasured states estimated by the model and do not correspond to the observed geographic distribution of species.

Region/parameter	Hidden Class Rate A		Hidden Class Rate B		Speciation (λ)
	Turnover (τ)	Extinction fraction (ϵ)	Turnover (τ)	Extinction fraction (ϵ)	
NA Endemic	0.45	0.06	5.2×10^{-9}	0.06	0.234
SA Endemic	0.05	0.06	0.46	0.06	0.314
Widespread	0.48	—	11.73	—	0.334

Discussion

This study sheds new light on the historical biogeography of Icteridae, a widespread and ecologically diverse New World avian clade. By combining phylogenetic bioregionalization, probabilistic inference of historical range evolution, and geography-dependent diversification modeling, this study offers an integrated view of how biogeographic events shaped the evolutionary history of this family. The findings support a Nearctic origin for Icteridae followed by a progressive southward expansion and *in situ* diversification across the Americas, refining and expanding upon prior hypotheses about the evolutionary trajectory of the family and offering broader insights into the macroevolutionary mechanisms shaping New World avifauna.

The ancestral area reconstructions strongly support a Nearctic origin for Icteridae, corroborating the molecular phylogenetic evidence of Barker et al. (2015) and refuting earlier ideas of a Neotropical origin put forward by Mayr (1946). The reconstructed colonization sequence, from the Nearctic into Mesoamerica (~7.53 Ma), then to Amazonia (~6.06 Ma), the Andes (~5.84 Ma), Southern South America (~4.83 Ma), and finally Patagonia (~2.68 Ma), traces a stepwise southward progression consistent with the gradual DEC model that provided the best fit to the data. This pattern is consistent with broader trends identified across New World birds, where avian lineages with Nearctic origins have repeatedly dispersed into the tropics and radiated throughout South America, while lineages of South American tropical origin remain largely restricted to the Neotropical region (Smith and Klicka 2010). The asymmetry in dispersal directionality recovered for Icteridae, therefore, could reflect a broader macroevolutionary pattern of niche conservatism and southward ecological expansion that has shaped New World avifauna more generally (Hawkins et al. 2006; Smith and Klicka 2010; Smith et al. 2012).

Concordantly, most of the early speciation events in Icteridae occurred during the mid- to late Pliocene, coinciding with the progressive expansion of open habitats (Anderson 2006; Jaramillo 2023), which appears to be the ancestral habitat for the family (Suppl. material 1: fig. S5). This result is consistent with open habitat being the predominant habitat association among lineages at the base of the tree, including the meadowlark and allies clade (*Xanthocephalus*, *Dolichonyx*, *Sturnella*, and *Leistes*), which is sister to

all other Icteridae. Further expansion of these habitats and their displacement of woody biomes as the planet became drier (Pagani et al. 1999; Osborne 2008) might have driven the expansion and dispersal of Icteridae, as its lineages were following their ancestral niche in open grass-dominated biomes (Wiens and Donoghue 2004; Crisp et al. 2009) or the retreating woody biome (Jaramillo 2023).

As such, the timing of the earliest dispersal from North America to South America (~6.06 Ma) predates the traditionally accepted establishment of the Isthmus of Panama (between ~3 and 2.7 Ma; Coates and Stallard 2013; Bourke et al. 2025), suggesting that early colonization of South America occurred via overwater dispersal or through pre- or proto-isthmian land connections. This timing agrees with the expansion of open habitats in northern South America due to Andean uplift approximately 9–6 Ma (Kirschner and Hoorn 2020). Moreover, the timing also coincides with growing biological evidence that the Great American Biotic Interchange (GABI) was a long and complex process beginning as early as the Oligocene–Miocene transition, with significant dispersal waves of terrestrial organisms documented at approximately 20 and 6 Ma (Bacon et al. 2015). While following the suggested pattern of northern colonization with subsequent speciation in South America (Smith and Klicka 2010), such events occurred at least 3 million years earlier than previously suggested, following more closely the timing of a dispersal wave (Bacon et al. 2015) than continuous dispersals after or during the establishment of the Isthmus ~3 Ma (Weir et al. 2009; Smith and Klicka 2010; Barker et al. 2015). This early dispersal of Icteridae to South America coincides with recent fossil evidence from North American mammals, which accumulated in Mesoamerica through the Miocene before a gradual exchange with South America approximately 7–3 Ma, also driven by the expansion of open habitats (Tseng et al. 2026).

Mesoamerica was both the first Neotropical region colonized from the Nearctic and the highest net source of dispersal events across the entire history of the clade, while simultaneously exhibiting the greatest number of range contractions and *in situ* speciation events among all regions. This combination of high dispersal, high speciation, and high lineage turnover suggests that Mesoamerica has acted as a highly dynamic biogeographic hub rather than a simple stepping-stone. As such, Mesoamerica harbors exceptionally high avian endemism generated by a complex

geological and ecological history (Rocha-Méndez et al. 2019) and functions as a key relay in biotic interchange, where topographic heterogeneity promotes both diversification and lineage turnover (McCormack et al. 2023). The peak in range contraction counts in Mesoamerica approximately 3 Ma (Suppl. material 1: fig. S4) may also reflect competitive dynamics following land-bridge establishment, as newly arrived lineages displaced earlier colonists (Weir et al. 2009; Smith and Klicka 2010). At the same time, most *in situ* speciation events in the region occurred during the Pleistocene. Examples of these events include most *Quiscalus* and *Icterus* lineages, which had already begun to diverge during the Pliocene (Powell et al. 2014). The timing of these cladogenetic events may be related to rapid diversification driven by the characteristic climatic oscillations of this epoch, which caused extensive habitat fragmentation and ultimately allopatric speciation of already diverging lineages (Avice and Walker 1998; Avice et al. 1998; Weir 2006). In fact, such oscillations are well documented to isolate avian populations and generate the allopatric conditions necessary for speciation in Mesoamerica, particularly in open habitats (Barber and Klicka 2010; Castillo-Chora et al. 2021).

Within South America, the Amazonia region functioned as the primary continental entry point and dispersal hub to Southern South America even early on, with the second-highest mean *in situ* speciation after Mesoamerica. Its role as both a major source of dispersal events and a center of *in situ* speciation agrees with the general pattern documented across many other Neotropical clades, for which Amazonia is recognized as the primary source of Neotropical biodiversity, providing over 2,800 lineages to other regions (Antonelli et al. 2018). The Andes region seems to have functioned primarily as a dispersal corridor rather than a center of diversification, as it was both a source and recipient of comparable dispersal events to different South American regions (Suppl. material 1: table S8), contrasting with the well-documented role of the Andes as a diversification hotspot for other Neotropical avian clades, such as thraupids and New World sparrows, where topographic complexity and elevational barriers have promoted allopatric speciation (Hazzi et al. 2018). One possible explanation for this contrast is that, unlike montane-endemic radiations, avian lineages associated with open habitats, such as early icterids, have higher dispersal ability (Sheard et al. 2020) and thus may be more likely to use the Andes as a connective corridor than as an isolating barrier (Dauerman et al. 2026).

Colonization proceeded southward, with Amazonia contributing substantially to the assembly of the Southern South American fauna. *In situ* speciation events are relatively few in Southern South America, consistent with biogeographic patterns in other New World birds (Smith and Klicka 2010). With respect to Icteridae, Southern South America was a net recipient rather than a net source, likely reflecting the non-Neotropical origin of the clade. Interestingly, South American regions exhibited early *in situ* diversification during the Pliocene, coinciding with the firm establishment of open habitats (such as páramo, floodplains, and savannas) in South America (Kirschner and Hoorn 2020; Jaramillo 2023).

Patagonia stands out across all biogeographic metrics as a very distinct region. It received dispersal almost exclusively from Southern South America, harbored virtually no *in situ* speciation, and showed the lowest range contraction counts. These features seem consistent with a region that might have been colonized late, rarely, and sparsely (Vuilleumier 1991; Bukowski et al. 2024). The nearly absent speciation signal is consistent with recent findings that colonization of new regions does not automatically generate elevated diversification (Kennedy et al. 2017; Arango et al. 2025).

Beyond the family-wide pattern described above, the major subclades of Icteridae differ notably in their individual biogeographic trajectories (see Suppl. material 1 for a full discussion). The meadowlarks and allies largely mirror the family-wide pattern, diversifying in North America during the late Miocene while retaining their ancestral open-habitat niche, with range expansion rather than range shift driving their spread into South America. The caciques, oropendolas, and allies show the opposite habitat trajectory, tracking retreating closed-forest habitats as they expanded from a North American origin into Mesoamerica and Amazonia, where speciation increased. The orioles diversified extensively within Mesoamerica during the Pliocene, with subsequent dispersal to South America during the Pleistocene. Furthermore, this group accounts for most reversals of the broader southward colonization trend of the family, with several South American lineages returning to North America (Winger et al. 2014). Finally, the grackles and allies split into two contrasting patterns: one lineage remained largely confined to Mesoamerica and North America until the Pleistocene, and a second radiated into Amazonia by the Miocene–Pliocene boundary alongside the expansion of open habitats (Jaramillo 2023). Together, these subclade-level differences suggest that the overall stepwise, unidirectional colonization pattern of Icteridae potentially conceals some heterogeneity in the timing, direction, and ecological correlates of range evolution among its major lineages.

Finally, the macroevolutionary analyses indicate that the full GeoHiSSE model provided the best fit, revealing that diversification dynamics in Icteridae are shaped by both observed geographic occupancy and unmeasured hidden states. This implies that geography alone does not fully capture the heterogeneity in speciation and extinction rates across the clade (Caetano et al. 2018).

The two hidden rate classes reveal distinct macroevolutionary regimes that are consistent with the biogeographic history recovered by the BSM analyses. In hidden rate class A, cladogenetic rates are moderate and geographically structured. NA endemics show the highest within-area speciation, consistent with the deep early diversification recovered in the Nearctic by the BSMs, while SA endemics show the lowest values with a non-negligible extinction fraction, pointing to a more dynamic macroevolutionary regime in the Neotropics, where speciation is offset by extinction, as documented across other South American passerine radiations (Ricklefs 2006b). As such, the elevated extinction fraction relative to its turnover among SA icterid endemics in hidden rate class A is consistent with the high range contraction rates recovered

by the BSMs in Andean and Amazonian regions, suggesting that lineage-level turnover has been a persistent feature of the history of Icteridae in South America.

WS lineages in hidden rate class A show high cladogenetic rates, consistent with the ongoing intercontinental exchange documented in the BSMs and with the high dispersal ability and migratory behavior that characterize many icterid lineages (Kondo and Omland 2007; Price et al. 2009; Arango et al. 2022). This elevated rate mirrors the high within-area speciation observed in NA endemics under the same hidden rate class, suggesting that the underlying hidden state continues to promote cladogenesis in WS lineages even as they expand into South America. This pattern suggests that the state has a similar effect in North America alone and in lineages spanning North and South America and, as such, might reflect a process of biome tracking in which early icterid species dispersed into South America as open habitats expanded during the Pliocene (Anderson 2006; Kirschner and Hoorn 2020; Jaramillo 2023). This interpretation is consistent with the broader pattern of niche conservatism documented across icterid lineages (Eaton et al. 2008) and the timing of the transition of icterid lineages into South America (Fig. 1b).

In hidden rate class B, the relationship between geography and diversification is qualitatively different and decoupled from the pattern in class A. NA endemics show near-zero cladogenetic rates, while SA lineages show rates comparable to those of class A NA lineages and WS lineages show the highest cladogenetic rate of any state combination. Due to the parameterization, turnover in WS lineages reflects rapid cladogenesis across all three ranges at once rather than a conventional high-turnover cycle of speciation and extinction (Goldberg et al. 2011; Caetano et al. 2018). These lineages therefore split rapidly into endemic daughter lineages on each continent, a pattern that, at the intercontinental scale, mirrors the dominance of cladogenetic range splitting documented across emberizoid passerines (Arango et al. 2025).

This contrast in cladogenetic rates depending on hidden states demonstrates that geography does not drive diversification in isolation. Instead, the same geographic area yields distinct diversification rates depending on the hidden regime of the lineage, indicating that geography merely contributes to a more complex rate structure. Importantly, the near-zero subsequent diversification of North American endemics in hidden rate class B does not reflect a geographic constraint on diversification in North America *per se*, as the Nearctic clearly supported substantial diversification in rate class A. This pattern indicates that lineages in hidden rate class B that happen to become NA endemics do not subsequently radiate (Caetano et al. 2018).

As such, mean speciation rates were comparable between continents, suggesting that both regions contributed substantially to overall clade richness, while the rate of lineage splitting in Icteridae is predominantly governed by an unmeasured biological axis that does not covary with either continent or range size. This finding resonates strongly with recent work on the broader Emberizoidea superfamily, which showed that broad geographic dispersal is not a

diversification driver for the group and that cladogenetic events occur primarily within bioregions rather than between them, suggesting that ecological opportunity, habitat specialization, and interspecific relationships within cohesive bioregions drive speciation more than dispersal events between regions (Arango et al. 2025). Specifically, ecological specialization has been identified as a significant positive predictor of speciation rates across Emberizoidea at continental scales, including in Icteridae (Conway and Olsen 2019), while dispersal ability shows no significant relationship with speciation rates in the clade (Arango et al. 2026). Moreover, sexual selection through plumage and song divergence is recognized as a critical process for speciation (Higashi et al. 1999). In Icteridae and allied groups in particular, color evolution and dichromatism have been shown to potentially drive species cladogenesis (Friedman et al. 2009; Price and Eaton 2014). These lines of evidence point toward ecologically driven within-region speciation that could be mediated by feeding specialization, social behavior, and sexual selection as the most likely biological axes underlying the hidden diversification regimes recovered by GeoHiSSE, rather than any geographically structured process. Formally testing this hypothesis will require integrating these traits directly into future diversification analyses of Icteridae.

Taken together, these results suggest that the diversification history of Icteridae was shaped by the interplay between geography-mediated colonization, regional ecological opportunity, and different macroevolutionary regimes, with the Nearctic providing an ancient source of deep early diversification (Arango et al. 2025); Mesoamerica acting as a dynamic filter, relay, and site of Plio-Pleistocene *in situ* diversification (Barber and Klicka 2010; Castillo-Chora et al. 2021); and the core Neotropical regions accumulating diversity through a combination of immigration and *in situ* speciation (Smith and Klicka 2010; Harvey et al. 2020). Meanwhile, the macroevolutionary results establish that speciation rate is primarily governed by factors other than geography, with a secondary geographic component that is consistent with—but does not strongly drive—the continental patterns recovered by the BSMs. These results not only corroborate the origin of Icteridae posited by Barker et al. (2015) but also support the stepwise colonization hypothesis of Mayr (1946), although the findings indicate that colonization occurred in the reverse direction.

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Additional information

Conflict of interest

The authors have declared that no competing interests exist.

Ethical statement

No ethical statement was reported.

Artificial Intelligence (AI) use

The authors accept full responsibility for the content of the manuscript, including the disclosure of any use of AI.

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Author contributions

Axel Arango: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Writing—original draft (primary); Writing—review and editing (primary). Chaitanya S. Gokhale: Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing—review and editing (secondary).

Author ORCIDs

A. Arango  <https://orcid.org/0000-0001-6246-8141>

C.S. Gokhale  <https://orcid.org/0000-0002-5749-3665>

Data availability

Data and code used to conduct all analyses presented are available on Figshare at <https://doi.org/10.6084/m9.figshare.33145841> and can be freely accessed.

Supplementary materials

Supplementary material 1

Additional figures and tables (.docx)

Link: <https://doi.org/10.21425/fob.19.202331.suppl1>